



Study to help shape policies for LEDs, solar panels, and batteries

Researchers at University of Cambridge (England) and ETH Zurich (Switzerland) studied their 30-year evolution, noting cost drops from economies of scale and efficiency gains from



These bulbs use microscopic LED chips, covered in a yellow phosphor substance to mimic a retro aesthetic while achieving high energy efficiency in lighting (Credit: Rambling Geek (<https://500px.com/p/ramblinggeek/>); WikiMedia Commons)

R&D. ‘Technology spillovers’ also boosted light quality, aiding adoption. The study recommends user-focused innovation and the effective use of external knowledge. Findings will help shape policies for LEDs, solar panels, and batteries, with improved methods for forecasting future technology costs.

Vapour deposition method to create better transistors

Researchers at Pohang University of Science and Technology (South Korea) developed a vapour deposition method using PbCl_2 , SnI_2 , and CsI to create tin-halide perovskite TFTs. This technique forms stable cesium-tin-iodide films with high hole mobility ($33.8\text{cm}^2/\text{V}\cdot\text{s}$) and excellent on/off ratios ($\sim 10^8$). Compared to solution methods, it offers better performance and scalability. These p-type transistors improve OLED efficiency and reduce power use, paving the way for large-area, low-cost electronics and advanced stacked circuit designs.

New 3D display panels improve human-computer interaction and safety

Scientists at the University of Science and Technology of China developed a 3D display panel using circularly polarised luminescence (CPL) technology. This display features an orthogonal-CPL emission mechanism, providing clearer stereoscopic images with broader viewing angles and reduced eye strain. The benefit is real-time depth perception and interactive hand gesture control, creating immersive experiences. The advantage lies in its wide applications—from industrial automation to medical electronics—enhancing human-computer interaction and safety in complex tasks.

OLED display with immersive, crosstalk-free audio without external speakers

Researchers at POSTECH (Pohang, South Korea) have developed the world’s first pixel-based Local Sound OLED

technology, which was demonstrated on a 33cm (13-inch) OLED panel. This innovation integrates ultra-thin piezo-electric exciters into the display, enabling each pixel to emit distinct sounds simultaneously. The benefit is immersive, crosstalk-free audio without the need for external speakers. The advantage is a slim, high-fidelity display that supports multisensory experiences in devices like cars, VR headsets, and mobile gadgets, revolutionising multimedia interaction.

India builds fast-charging, low-cost sodium-ion battery

As India pushes for clean energy and self-reliance, researchers at JNCASR (Bengaluru, Karnataka), under DST, have developed a fast-charging sodium-ion battery (SIB) using a NASICON-type cathode and a novel anode: $\text{Na}_{1.0}\text{V}_{0.25}\text{Al}_{0.25}\text{Nb}_{1.5}(\text{PO}_4)_3$. It charges to 80% in just six



(left) Fast charging sodium ion battery; (right) Researchers of this discovery - Biplab Patra (PhD student, JNCASR) and Prof. Premkumar Senguttuvan, Associate Professor, JNCASR (Credit: <https://www.pib.gov.in>)

minutes and lasts over 3000 cycles. Innovations include nanoscale particle sizing, a carbon coating, and aluminium doping, enabling faster, safer ion movement. Sodium, unlike lithium, is abundant

and locally available, reducing import dependence. The SIB shows promise for EVs, solar grids, and rural electrification, as demonstrated through electrochemical cycling and quantum simulations for enhanced safety and performance.

Latch effect in GaN semiconductors benefits 6G technology development

Research at the University of Bristol (England) revealed a latch effect in gallium nitride (GaN) semiconductors, enabling faster, more powerful radio frequency amplifiers. This breakthrough, demonstrated in superlattice castellated field-



An artistic image of a futuristic semiconductor device which will help make 6G technology a reality (Credit: University of Bristol)

effect transistors with tiny fins, enhances device performance and reliability without harm. The advancement benefits 6G technology development, paving the way for innovations like self-driving cars, instant health diagnoses, and remote touch experiences. Industry collaboration will drive commercial use.